



MACHINE LEARNING CLASSIFICATION ASSIGNMENT

Breast Cancer Wisconsin Dataset



FEBRUARY 15, 2026

RAMAMURTHY S
2025ab05099

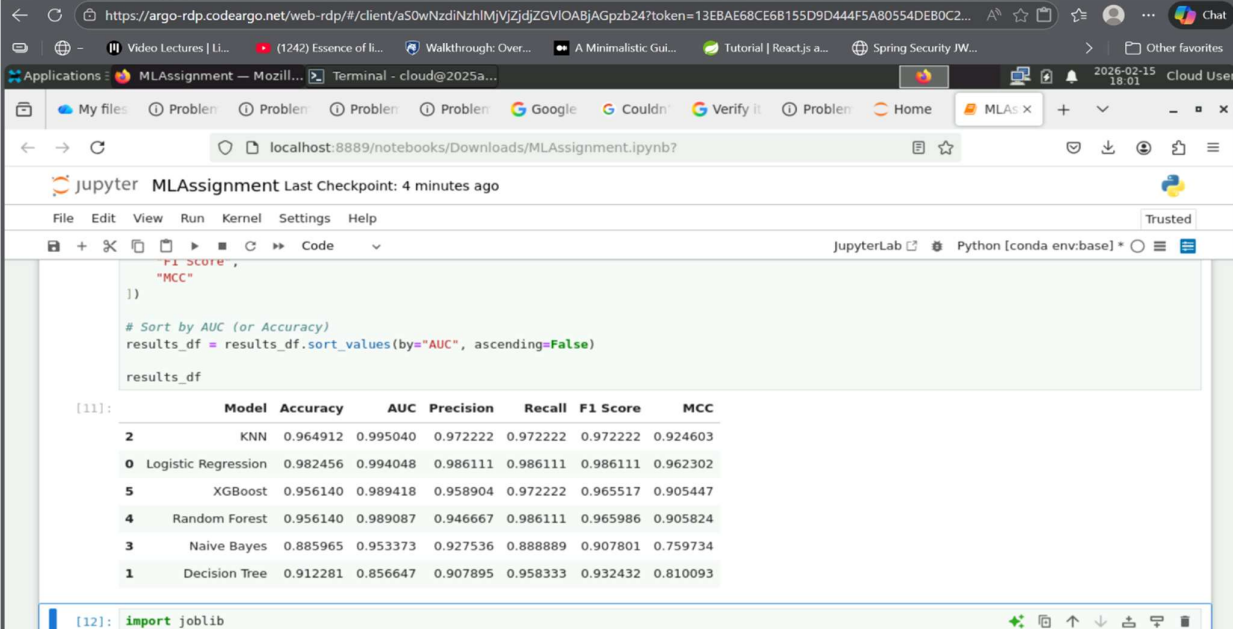
Git Hub Repository Link

<https://github.com/ram-2025ab05099/ml-classification-streamlit>

Live Streamlit App Link

<https://ml-classification-app-o8hfwmbzbugn5bvsp7brn2.streamlit.app/>

Screenshot – BITS Virtual Lab Execution



The screenshot shows a JupyterLab interface with a code cell containing the following code:

```
def f1_score,
"MCC"
})

# Sort by AUC (or Accuracy)
results_df = results_df.sort_values(by="AUC", ascending=False)
results_df
```

The output of the code cell is a table with 7 columns: Model, Accuracy, AUC, Precision, Recall, F1 Score, and MCC. The table contains 7 rows of data, sorted by AUC in descending order.

	Model	Accuracy	AUC	Precision	Recall	F1 Score	MCC
2	KNN	0.964912	0.995040	0.972222	0.972222	0.972222	0.924603
0	Logistic Regression	0.982456	0.994048	0.986111	0.986111	0.986111	0.962302
5	XGBoost	0.956140	0.989418	0.958904	0.972222	0.965517	0.905447
4	Random Forest	0.956140	0.989087	0.946667	0.986111	0.965986	0.905824
3	Naive Bayes	0.885965	0.953373	0.927536	0.888889	0.907801	0.759734
1	Decision Tree	0.912281	0.856647	0.907895	0.958333	0.932432	0.810093

The bottom of the screenshot shows a terminal window with the following code:

```
[12]: import joblib
import os
```

README Content

🧠 Breast Cancer Classification using Machine Learning

a. Problem Statement

The objective of this project is to build and compare multiple machine learning classification models to predict whether a breast tumor is ****malignant**** or ****benign**** using diagnostic medical measurements.

Six classification algorithms are implemented and evaluated using multiple performance metrics. The best-performing model is identified based on comparative analysis.

b. Dataset Description *(1 Mark)*

The dataset used in this project is the **Breast Cancer Wisconsin Dataset** from the UCI Machine Learning Repository.

📊 Dataset Characteristics

Total Instances: 569

Total Features: 30 numerical features

Target Classes:

- 0 → Malignant
- 1 → Benign

Feature Type: All numerical

Missing Values: None

The features are computed from digitized images of fine needle aspirate (FNA) of breast masses and describe characteristics such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension.

✓ Minimum 12 features satisfied

✓ Minimum 500 instances satisfied

✓ Binary classification problem

c. Models Used and Performance Comparison *(6 Marks)*

The following six machine learning classification models were implemented:

1. Logistic Regression
2. Decision Tree Classifier

- 3. K-Nearest Neighbors (kNN)
- 4. Gaussian Naive Bayes
- 5. Random Forest (Ensemble)
- 6. XGBoost (Ensemble)

All models were trained using an ****80:20 stratified train-test split****.

📊 Evaluation Metrics Used

- Accuracy
- AUC Score
- Precision
- Recall
- F1 Score
- Matthews Correlation Coefficient (MCC)

🔍 Model Comparison Table

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.982	0.994	0.986	0.986	0.986	0.962
Decision Tree	0.912	0.857	0.908	0.958	0.932	0.810
kNN	0.965	0.995	0.972	0.972	0.972	0.925
Naive Bayes	0.886	0.953	0.928	0.889	0.908	0.760
Random Forest (Ensemble)	0.956	0.989	0.947	0.986	0.966	0.906
XGBoost (Ensemble)	0.956	0.989	0.959	0.972	0.966	0.905

🚀 Observations on Model Performance *(3 Marks)*

ML Model Name	Observation about Model Performance
Logistic Regression	Achieved the highest accuracy and MCC, indicating the dataset is nearly linearly separable. It generalized extremely well with balanced precision and recall.
Decision Tree	Showed lower AUC compared to other models, likely due to overfitting and instability of a single tree model.
kNN	Performed very strongly with the highest AUC score, indicating good class separation based on distance metrics.
Naive Bayes	Lower performance due to strong feature correlations violating the independence assumption of the model.
Random Forest (Ensemble)	Performed very well with high recall and strong MCC. Ensemble learning improved robustness compared to a single decision tree.
XGBoost (Ensemble)	Comparable to Random Forest, achieved strong overall performance but did not significantly outperform Logistic Regression due to the small dataset size.

✅ Conclusion

****Logistic Regression emerged as the best-performing model****, suggesting that the dataset is largely linearly separable. While ensemble methods performed competitively, they did not significantly outperform simpler models due to the structured nature and relatively small size of the dataset.

🚀 Deployment

This project is deployed using ****Streamlit Community Cloud****.
To run locally:

```
`` `bash
pip install -r requirements.txt
streamlit run app.py
```

